

Real-Life Graphs

Read the axes first: what goes up, what goes across. The gradient is a rate; the area under a speed–time graph is the distance.



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THE TOPIC ON ONE MAP

Real-Life Graphs

01 Distance–time (travel) graphs

how long did he rest?

complete the return journey

who was faster?

02 Speed–time graphs

acceleration

area = distance

did Beth finish first?

03 Other real-life graphs

conversion graph

fixed charge + rate

interpret the gradient

TWO GRAPHS, TWO RULES Check which graph it is before you start:

ZONE DISTANCE--TIME

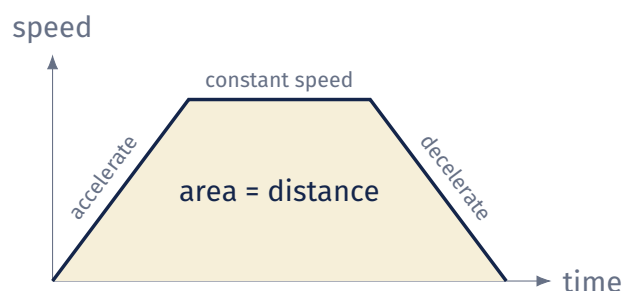
gradient = speed

flat = stopped; steeper = faster; back to 0 = home

ZONE SPEED--TIME

gradient = acceleration

area under the graph = distance; flat = constant speed



01 Distance–Time (Travel) Graphs

A flat line is a stop. The steeper the line, the faster. Speed = distance $\div$ time for each straight part — change minutes to hours for km/h.

WHEN YOU SEE → YOU DO

“how many minutes did he rest?”

→ length of the flat part

“complete the travel graph”

→ plot each new point (time, distance), join with straight lines

“who had the faster average speed?”

→ work out both speeds in the same units

“greatest speed — give a reason”

→ the steepest part

“average speed for the whole journey”

→ total distance $\div$ total time (stops included)

Read the scales

Distance $\div$ time

Minutes $\rightarrow$ hours

Compare / answer

WORKED MODEL • COACH

A coach travels 9 km in the first 10 minutes. Find its average speed in km/h.

1 10 minutes = $\frac{1}{6}$ hour

units

2 $9 \div \frac{1}{6}$

distance $\div$ time

ANSWER 54 km/h

× **TRAP** $9 \div 10 = 0.9$ “km/h” — the time was in minutes. Change to hours first.

IDEA 01

Practise: Distance–Time (Travel) Graphs

My questions first, then selected practice.

Q1 **DR ESLAM** Speed

CH 3 • TOPIC 7 2 marks

A cyclist travels 18 km in 45 minutes. Work out the average speed in km/h.

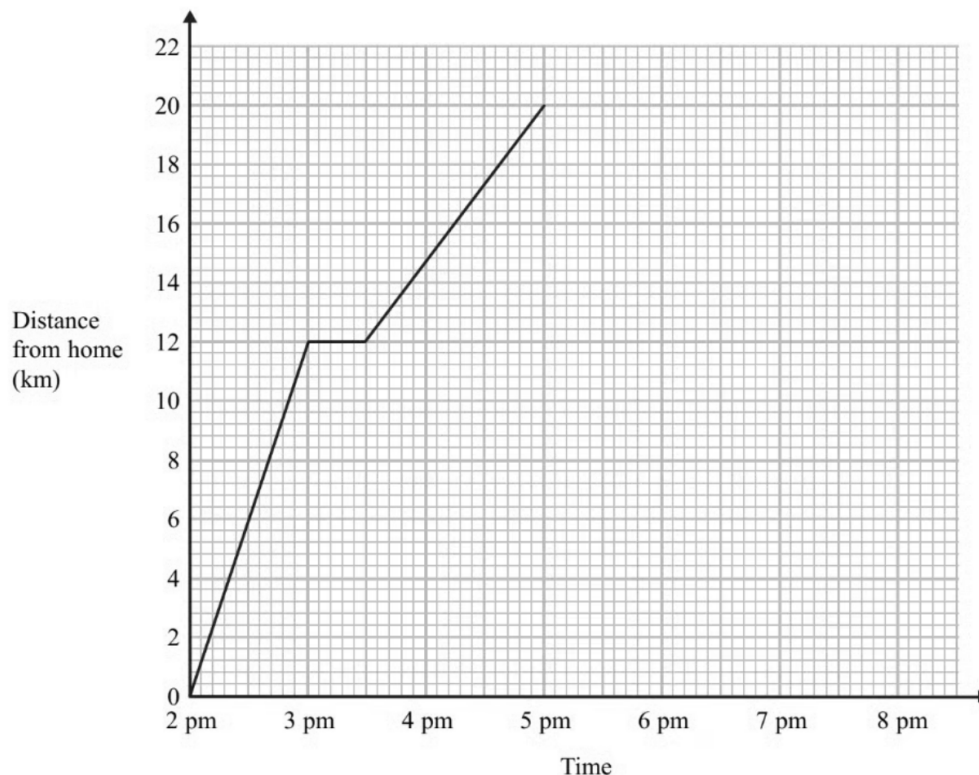
SELECTED PRACTICE

Q2**PRACTICE** Practice • Easy

CH 3 • TOPIC 7 4 marks

- 1 (a)** Simon went for a cycle ride.
He left home at 2 pm.

The travel graph represents part of Simon's cycle ride.



At 3 pm Simon stopped for a rest.

How many minutes did he rest?

(1 mark)

- (b)** How far was Simon from home at 5 pm?

(1 mark)

- (c)** At 5 pm Simon stopped for 30 minutes.
Then he cycled home at a steady speed.
It took him 1 hour 30 minutes to get home.

Complete the travel graph.

(2 marks)

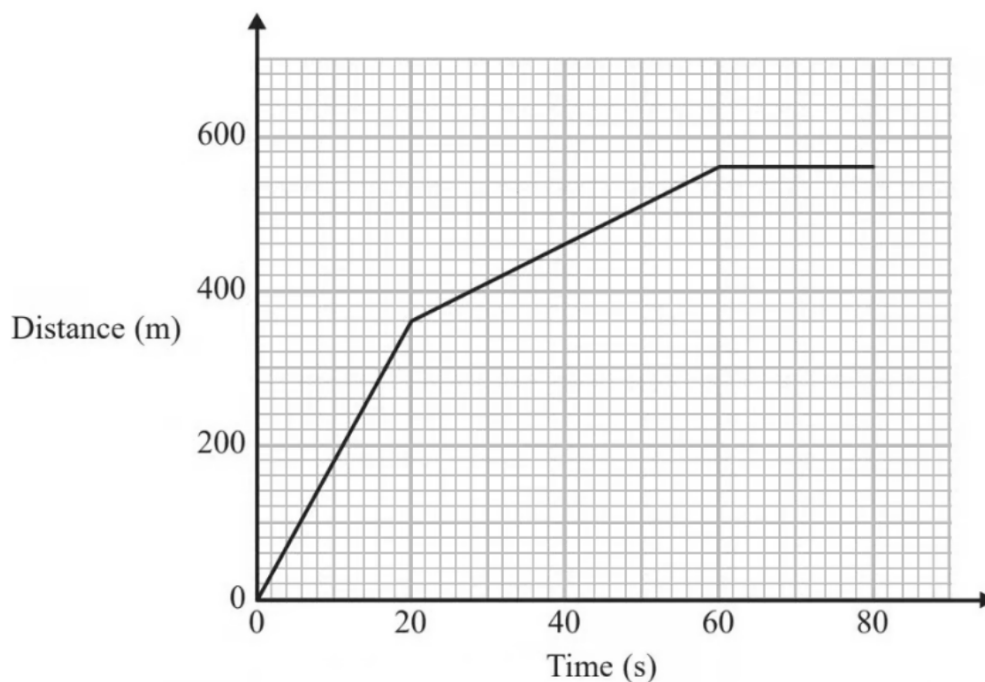
Q3**PRACTICE**

Practice • Easy

CH 3 • TOPIC 7

3 marks

3 (a) Here is part of a distance-time graph for a car's journey.



Between which two times does the car travel at its greatest speed?
Give a reason for your answer.

(2 marks)

(b) Work out this greatest speed.

(1 mark)

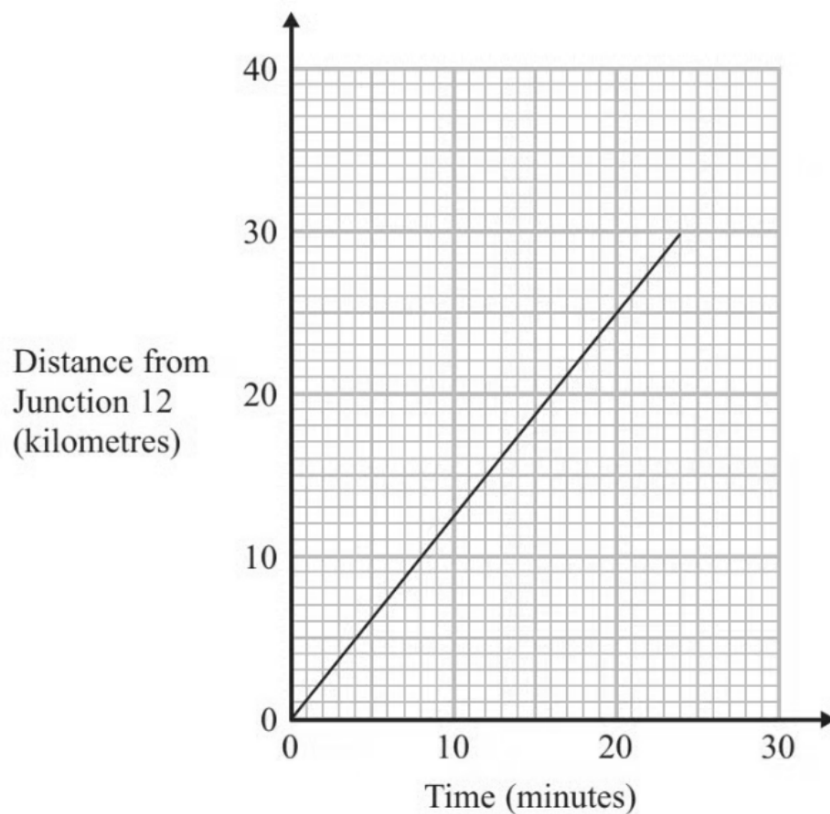
Q4**PRACTICE**

Practice • Medium

CH 3 • TOPIC 7 4 marks

- 2 Debbie drove from Junction 12 to Junction 13 on a motorway.

The travel graph shows Debbie's journey.



Ian also drove from Junction 12 to Junction 13 on the same motorway.
He drove at an average speed of 66 km/hour.

Who had the faster average speed, Debbie or Ian?
You must explain your answer.

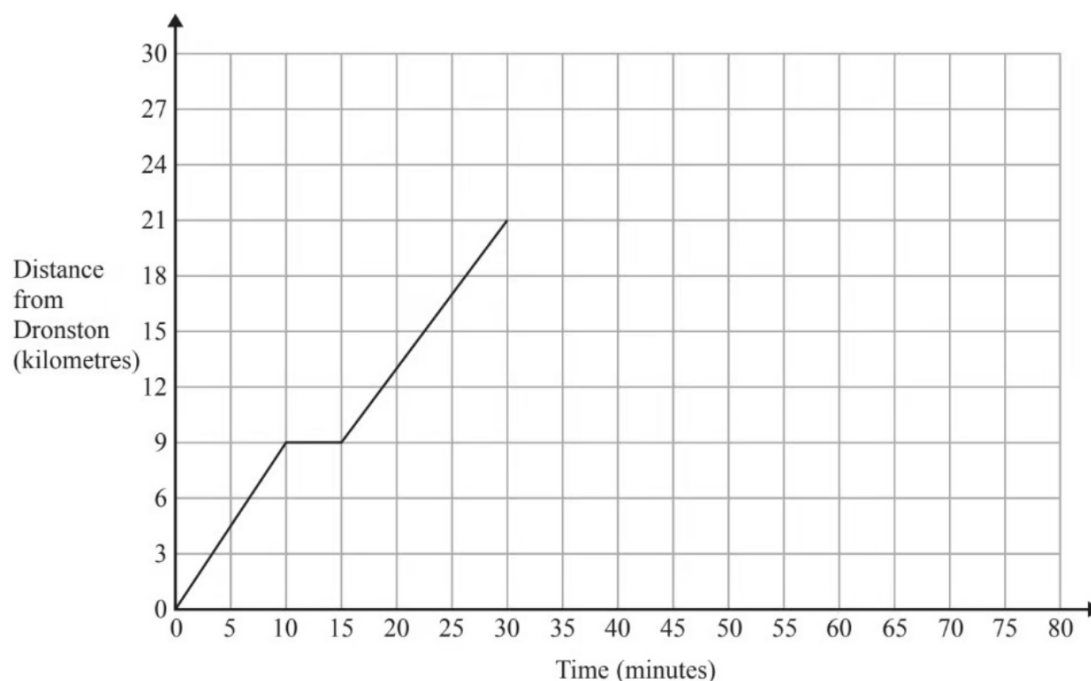
(4 marks)

Q5**PRACTICE**

Practice • Medium

CH 3 • TOPIC 7 5 marks

- 4 (a) A coach travels from Dronston to Luscoe.
The travel graph for this journey is shown below.



Work out the average speed of the coach, in kilometres per hour, for the first 10 minutes of the journey.

(2 marks)

- (b) The coach stops in Luscoe for 15 minutes.

The coach then returns to Dronston at a constant speed of 42 km/h.

Show this information on the travel graph.

(3 marks)

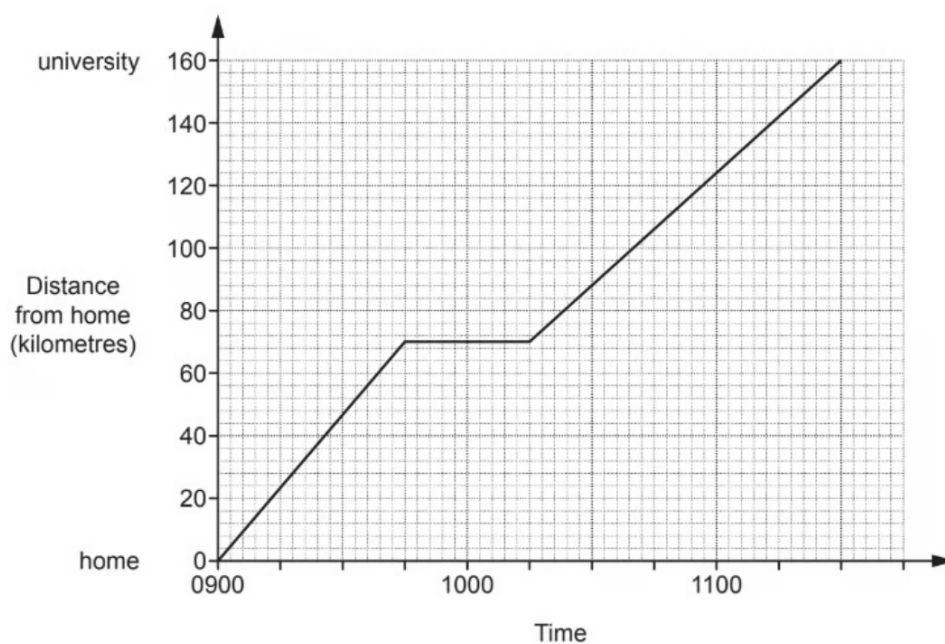
Q6

PRACTICE

Practice • Medium

CH 3 • TOPIC 7 3 marks

12 The graph shows Mia's journey from her home to university.



Calculate Mia's average speed for the whole journey.

..... km/h
(3 marks)

Q7**PRACTICE**

Practice • Hard

CH 3 • TOPIC 7

4 marks

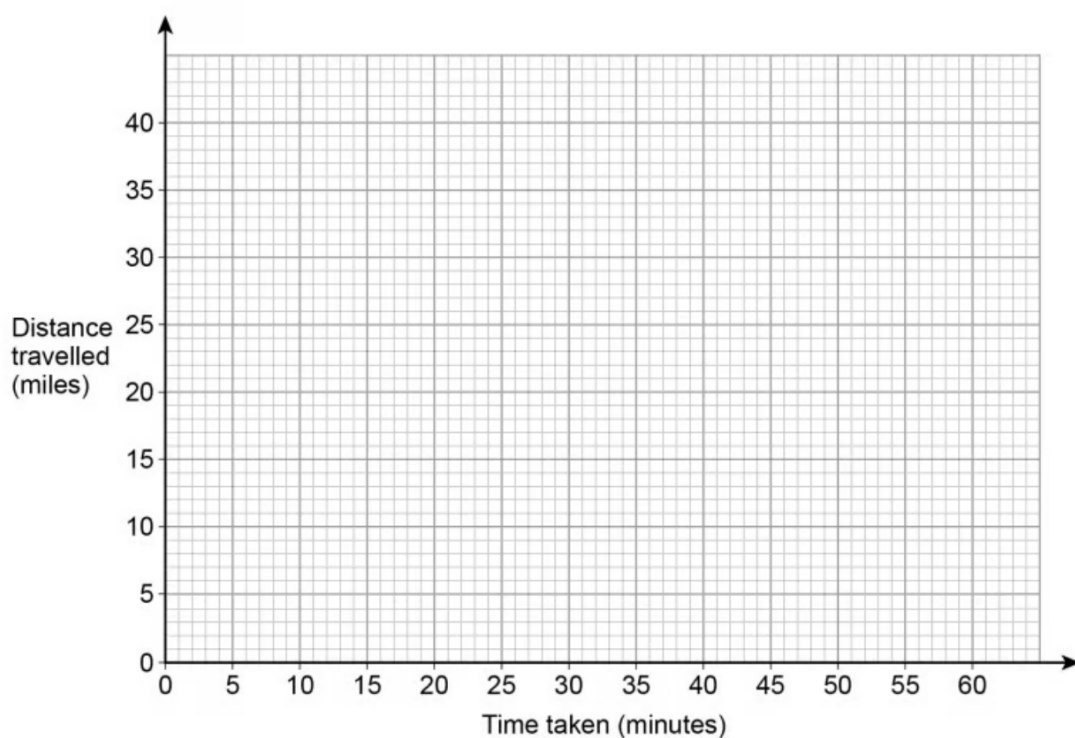
6 (a) Lily goes on a car journey.

For the first 30 minutes her average speed is 40 miles per hour. She then stops for 15 minutes.

She then completes the journey at an average speed of 60 miles per hour.

The total journey time is 1 hour.

Draw a distance-time graph for her journey.

**(3 marks)****(b)** Write down the average speed for the total journey.

..... mph

(1 mark)

02 Speed–Time Graphs

The gradient is the acceleration (negative = deceleration). The **area under the graph** is the distance travelled — split it into triangles and rectangles.

WHEN YOU SEE → YOU DO

“find the acceleration for the first 4 seconds”

→ gradient: change in speed ÷ time (m/s^2)

“distance travelled”

→ area under the graph

“a flat line”

→ constant speed, acceleration 0

“the car travelled 1.75 km — find V ”

→ area in terms of $V = 1750$

“average speed during the 60 seconds”

→ total area ÷ total time

WORKED MODEL • DISTANCE

Speed rises from 0 to 30 m/s in 15 s, then stays at 30 m/s until 40 s. Distance travelled?

1 triangle: $\frac{1}{2} \times 15 \times 30 = 225$

first part

2 rectangle: $25 \times 30 = 750$

second part

ANSWER 975 m

IDEA 02

Practise: Speed–Time Graphs

My questions first, then selected practice.

Q8 DR ESLAM Area

CH 3 • TOPIC 7 3 marks

A car accelerates from rest to 20 m/s in 10 s, then travels at 20 m/s for 30 s. Find the total distance.

SELECTED PRACTICE

Q9

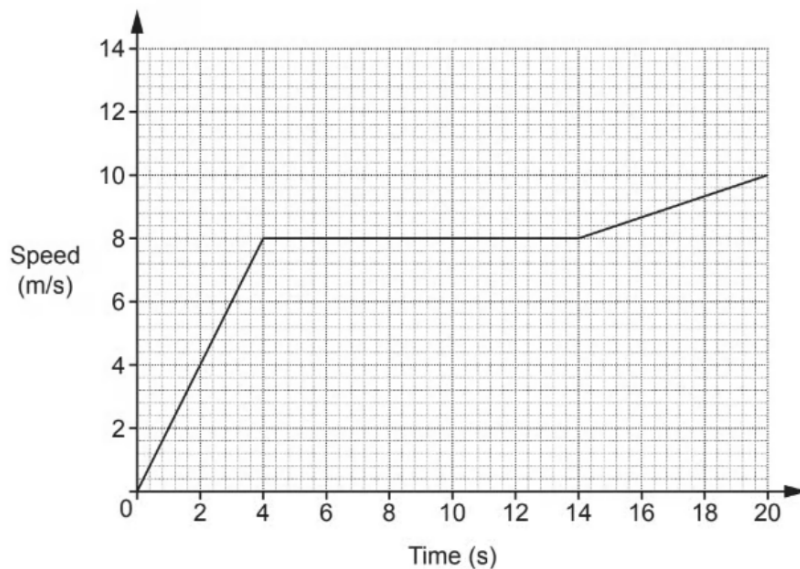
PRACTICE

Practice • Easy

CH 3 • TOPIC 7

3 marks

8 The graph shows the speed of a cyclist during 20 seconds of a journey.



Find the acceleration of the cyclist

(i) for the first 4 seconds

.....m/s² [2]

(ii) between 4 seconds and 14 seconds.

.....m/s² [1]

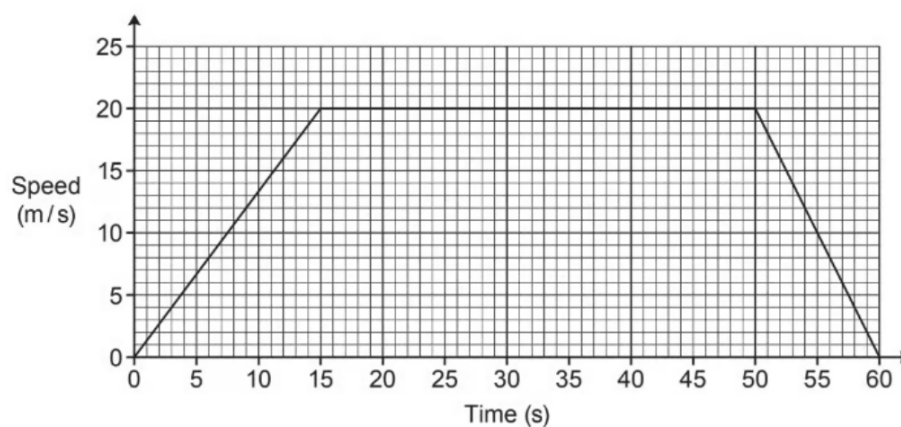
(3 marks)

Q10**PRACTICE**

Practice · Medium

CH 3 · TOPIC 7 7 marks

10 (a) The graph shows the speed of a train during the first 60 seconds of motion.



What is the speed of the train after 9 seconds?

..... m/s

(1 mark)

(b) What does the straight line suggest about the speed of the train over the first 15 seconds?

(1 mark)

(c) Work out the average speed of the train, in m/ s, during the 60 seconds.

..... m/ s

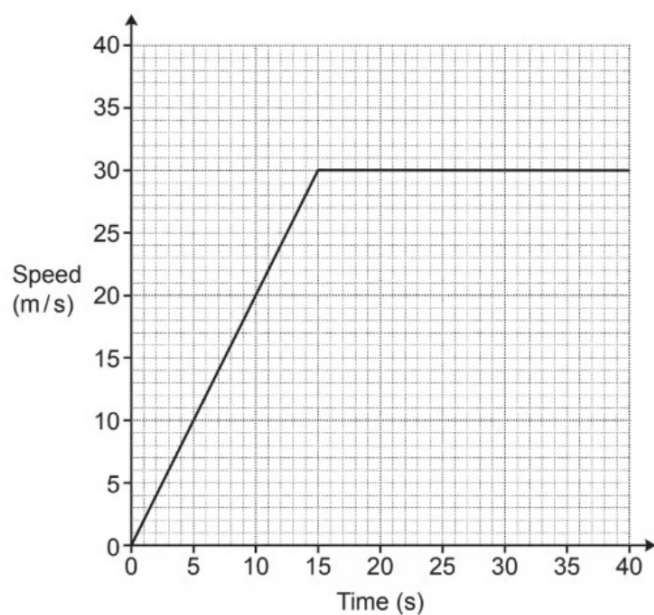
(5 marks)

Q11**PRACTICE**

Practice • Medium

CH 3 • TOPIC 7 **3 marks**

14 The graph shows the speed of a vehicle during the first 40 seconds of motion.



Calculate the distance travelled by the vehicle during the 40 seconds.

..... m

(3 marks)

Q12**PRACTICE**

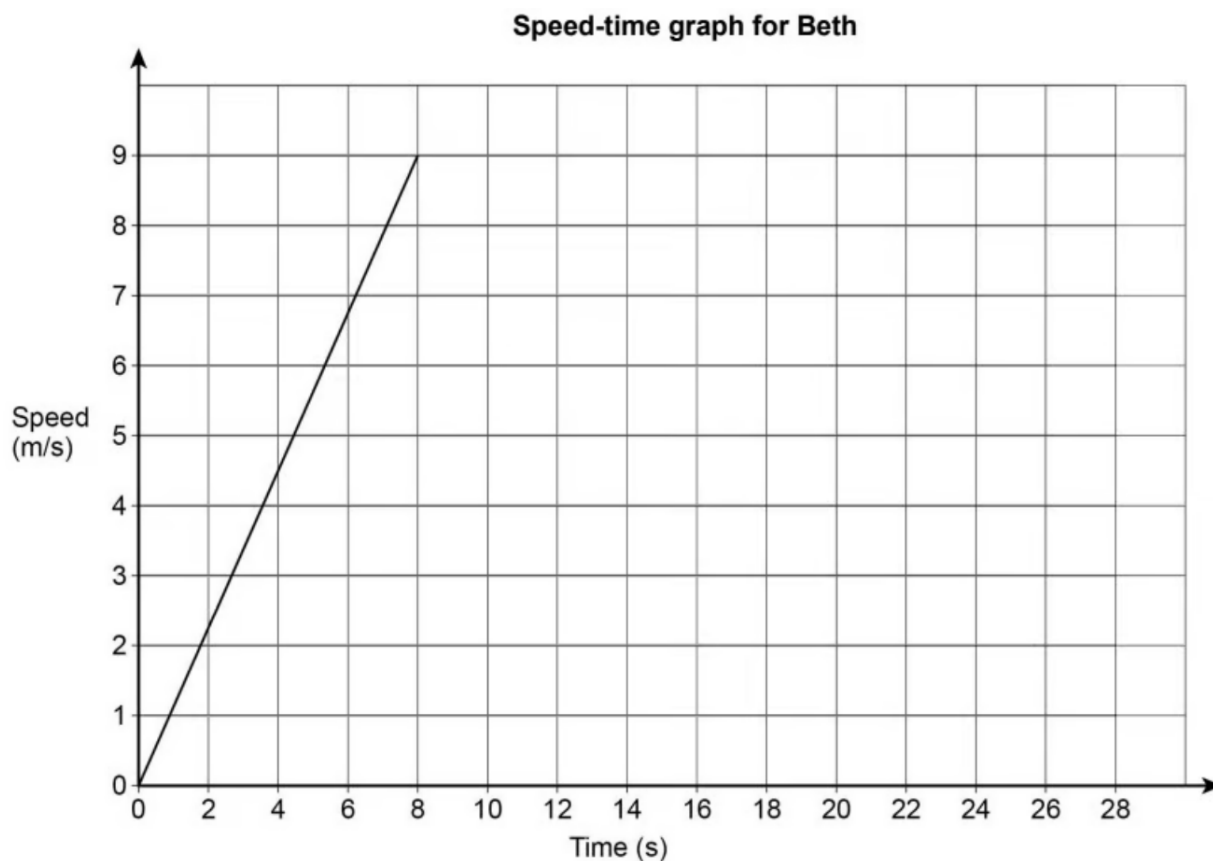
Practice • Hard

CH 3 • TOPIC 7 3 marks

5 Beth ran a 200 metre race.

Here is a graph of the first 8 seconds of her race.

She completed the race at a constant speed of 9 m/s



Amy completed the race in 27 seconds.

Did Beth finish before Amy?

You **must** show your working.

(3 marks)

Q13**PRACTICE**

Practice • Hard

CH 3 • TOPIC 7 6 marks

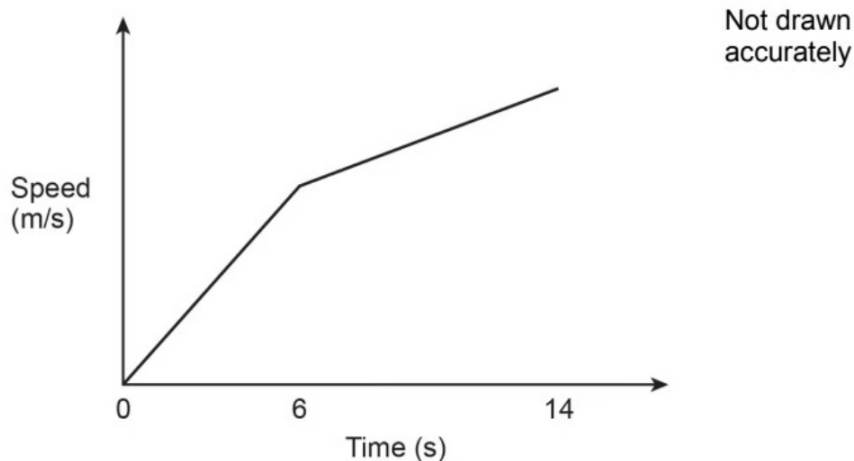
10 (a) Izzy runs an 80-metre race in 14 seconds.

During the first 6 seconds her speed increases at a constant rate.

During the last 8 seconds her speed increases at a different constant rate.

Her speed at 14 seconds is 2 m/s more than her speed at 6 seconds.

Here is a sketch of her speed-time graph.



Work out her acceleration during the last 8 seconds.

State the units of your answer.

(2 marks)

(b) When Izzy finishes the 80-metre race, her speed is v m/s

Work out the value of v .

(4 marks)

Q14**PRACTICE**

Practice • Hard

CH 3 • TOPIC 7 6 marks

13 (a) A toy car is placed on the floor of a sports hall.

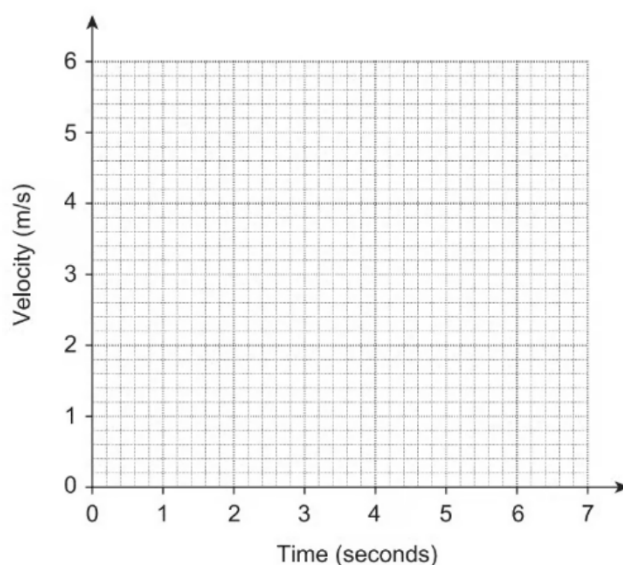
It moves in a straight line starting from rest.

It travels with constant acceleration for 4 seconds reaching a velocity of 5 m/s.

It then slows down with constant deceleration of 1 m/s^2 for 2 seconds.

It then hits a wall and stops.

Draw a velocity-time graph for the toy car.

**(3 marks)**

(b) Work out the total distance travelled by the toy car.

..... m

(3 marks)

03 Other Real-Life Graphs

Any straight-line graph is $y = mx + c$: c is the fixed or starting amount, m is the rate per unit. A conversion graph goes through O .

WHEN YOU SEE → YOU DO

“use the conversion graph to change £5 to dollars”

→ up from 5, across to the axis

a big amount off the graph (£800)

→ read a smaller value, then multiply

“how much is the fixed charge?”

→ the y -intercept

“what does the gradient represent?”

→ “litres per second”, “cost per mile”

“compare Bill’s and Ed’s costs”

→ find where the lines cross; say which is cheaper each side

WORKED MODEL • COMPARE COSTS

Bill: £10 fixed + £1 per mile. Ed: £1.50 per mile, no fixed charge. Compare.

1 $10 + x = 1.5x \Rightarrow x = 20$

where the lines cross

ANSWER Ed cheaper under 20 miles; Bill cheaper over 20 miles

IDEA 03

Practise: Other Real-Life Graphs

My questions first, then selected practice.

Q15**DR ESLAM**

Meaning

CH 3 • TOPIC 7

2 marks

A taxi costs £3 plus £1.20 per km. State the gradient and intercept of its cost graph and what they mean.

SELECTED PRACTICE

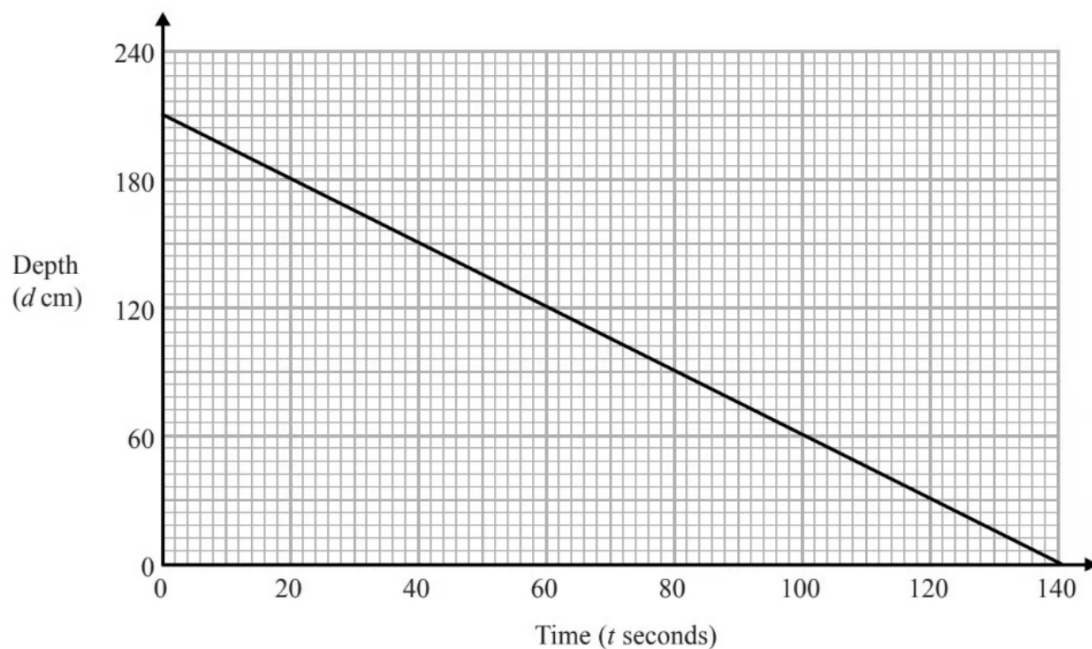
Q16

PRACTICE

Practice • Easy

CH 3 • TOPIC 7 3 marks

4 (a) The graph shows the depth, d cm, of water in a tank after t seconds.



Find the gradient of this graph.

(2 marks)

(b) Explain what this gradient represents.

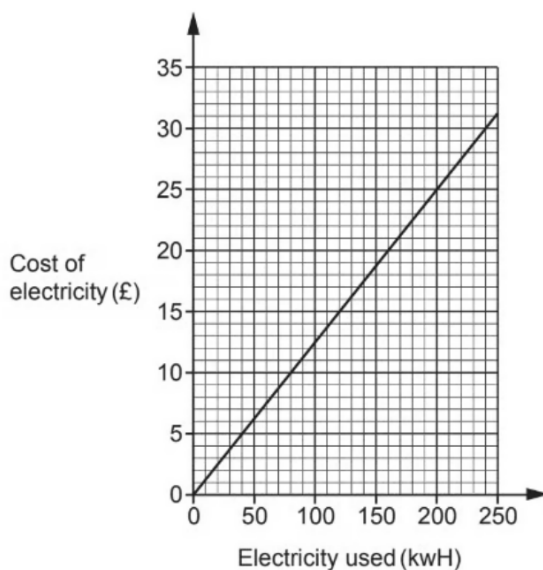
(1 mark)

Q17**PRACTICE**

Practice • Easy

CH 3 • TOPIC 7 6 marks

7 (a) The graph shows the cost of electricity with Company A.



Use the information in the graph to estimate the cost of electricity for a customer who uses 450kWh of electricity.

£

(3 marks)

(b) Company B charges 14.3 pence per kWh of electricity used.

If Company B's cost of electricity was plotted on the same axes as Company A's cost of electricity, which line would be steeper? Explain how you know.

Company would have the steeper line because

(3 marks)

Q18

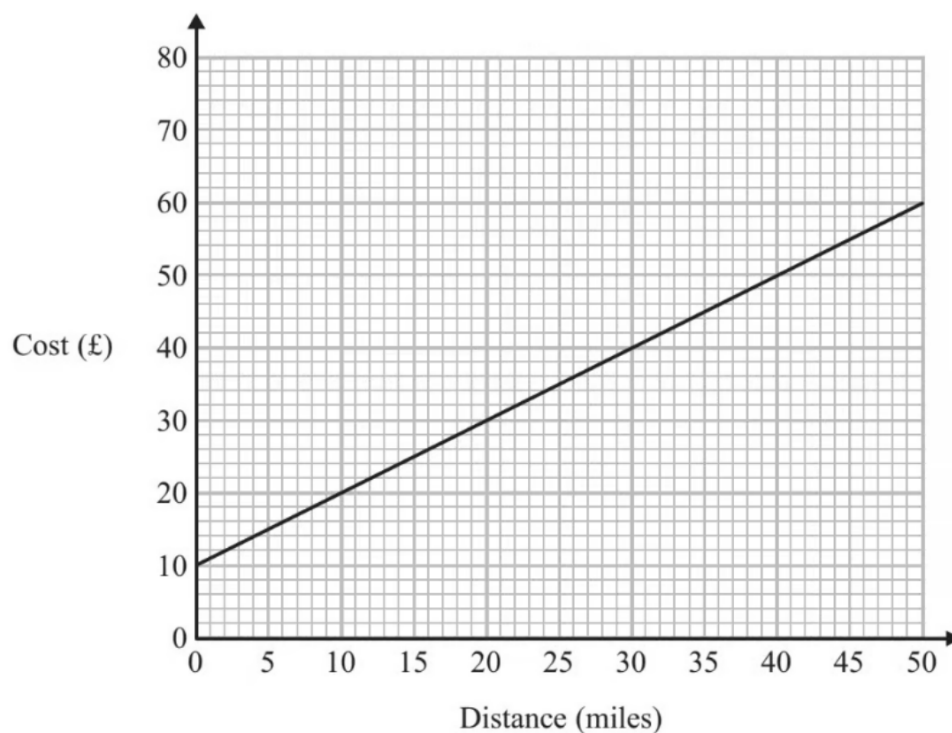
PRACTICE

Practice • Medium

CH 3 • TOPIC 7 4 marks

- 5 (a) Bill uses his van to deliver parcels.
For each parcel Bill delivers there is a fixed charge plus £1.00 for each mile.

You can use the graph to find the total cost of having a parcel delivered by Bill.



How much is the fixed charge?

(1 mark)

- (b) Ed uses a van to deliver parcels.
For each parcel Ed delivers it costs £1.50 for each mile.
There is **no** fixed charge.

Compare the cost of having a parcel delivered by Bill with the cost of having a parcel delivered by Ed.

(3 marks)

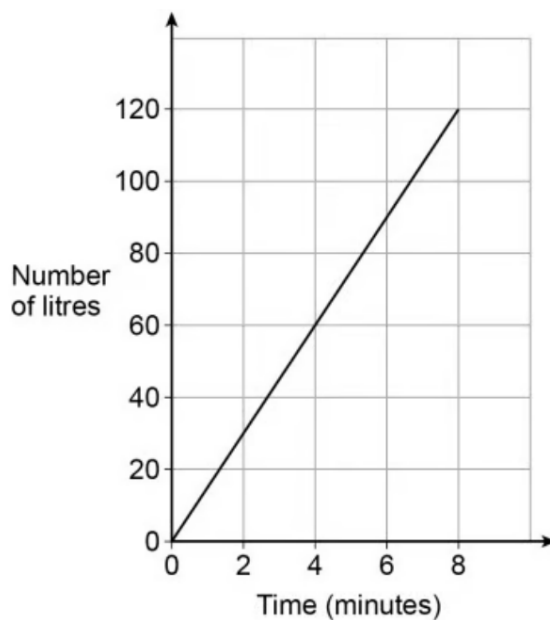
Q19**PRACTICE**

Practice • Medium

CH 3 • TOPIC 7 1 mark

9 Water is poured into a tank.

The graph shows the number of litres of water in the tank.



How much water is poured into the tank each minute? Circle your answer.

1.5 litres	15 litres	30 litres	120 litres
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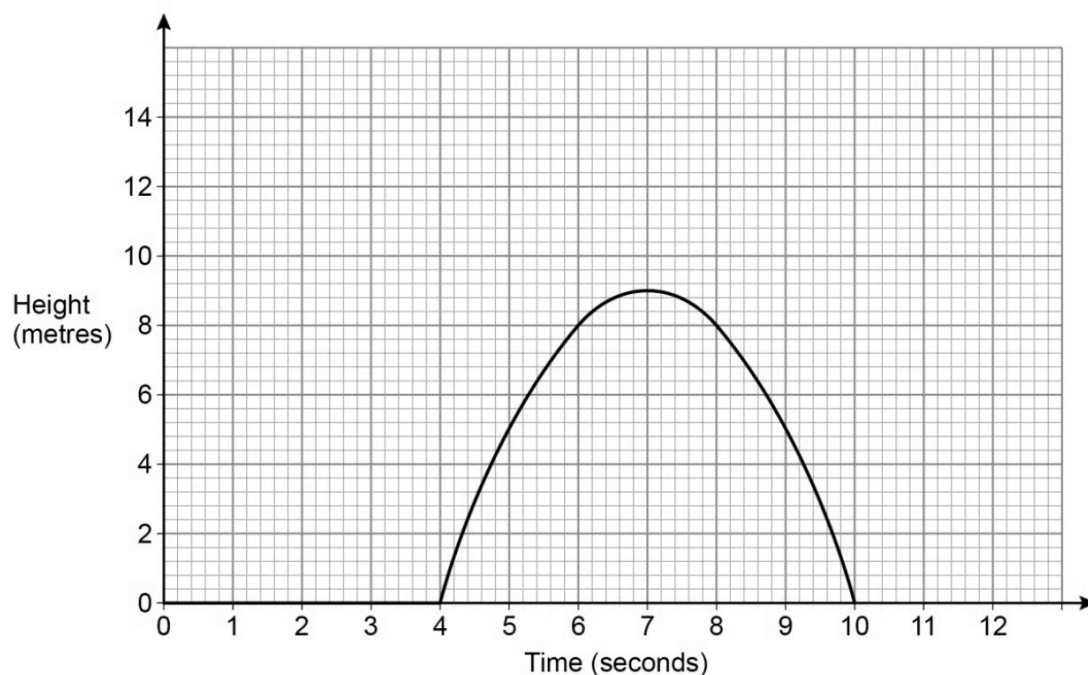
(1 mark)

Q20**PRACTICE**

Practice • Hard

CH 3 • TOPIC 7 4 marks

8 (a) The graph shows the height above ground of a toy rocket for 10 seconds.



For how long is the rocket in the air?

Circle your answer.

10 seconds	9 seconds	6 seconds	4 seconds
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(1 mark)

(b) Using the graph, estimate the speed of the rocket after 6 seconds. State the units of your answer.

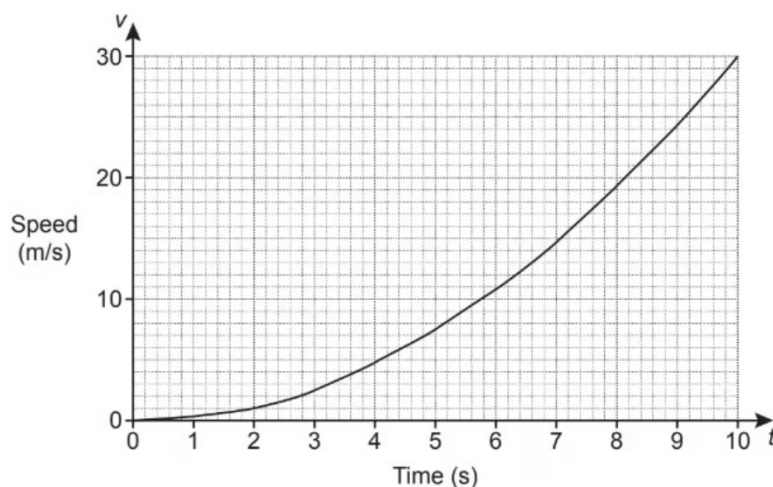
(3 marks)

Q21**PRACTICE**

Practice • Hard

CH 3 • TOPIC 7 10 marks

- 11 (a)** The graph shows the speed, v metres per second (m/s), of a car at time t seconds.



Find the speed of the car at $t = 7$.

..... m/s

(1 mark)

- (b)** It is claimed that the car has accelerated from 0 to 60 miles per hour in the first 10 seconds.

Does the graph support this claim?

Show your reasoning. Use 1 mile = 1.6 kilometres.

(5 marks)

- (c)** Use the graph to estimate the acceleration at $t = 7$.

.....m/s²

(3 marks)

- (d)** Georgina says that the graph shows that the speed of the car will continue to increase after 10 seconds.

Make one comment to show that this statement is incorrect.

(1 mark)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|----------------------|---|
| 01 | Distance–time | Gradient = speed; flat = stop; minutes $\rightarrow$ hours. |
| 02 | Speed–time | Gradient = acceleration; area = distance. |
| 03 | Other graphs | c = fixed / start amount; m = rate; say what it means. |

BOTTOM LINE Read the axes $\rightarrow$ gradient = rate $\rightarrow$ speed–time: area = distance.

Answers

Real-Life Graphs • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • DISTANCE--TIME (TRAVEL) GRAPHS

- Q1** Dr Eslam • Speed $18 \div 0.75 = 24 \text{ km/h}$
- Q2** Practice • Easy (a) 20 minutes (b) 20 km (c) flat 5:00–5:30 pm, line back to 0 at 7:00 pm
- Q3** Practice • Easy (a) 0–20 s: steepest line (b) 18 m/s
- Q4** Practice • Medium Debbie 30 km in 24 min = 75 km/h > 66: Debbie
- Q5** Practice • Medium (a) 54 km/h (b) flat 30–45 min, then back to 0 at 75 min
- Q6** Practice • Medium about 64 km/h (160 km in 2.5 h)
- Q7** Practice • Hard (a) (30, 20), flat to 45 min, (60, 35) (b) 35 mph

IDEA 02 • SPEED--TIME GRAPHS

- Q8** Dr Eslam • Area $100 + 600 = 700 \text{ m}$
- Q9** Practice • Easy (i) 2 m/s^2 (ii) 0
- Q10** Practice • Medium (a) 12 m/s (b) constant acceleration (c) $950 \text{ m} \div 60 \approx 15.8 \text{ m/s}$
- Q11** Practice • Medium 975 m
- Q12** Practice • Hard 36 m in 8 s, then $164 \div 9 = 18.2 \text{ s}$: $26.2 \text{ s} < 27$ — yes
- Q13** Practice • Hard (a) 0.25 m/s^2 (b) $11u + 8 = 80$, $u = \frac{72}{11}$: $v = \frac{94}{11} \approx 8.55$
- Q14** Practice • Hard (a) $(0, 0) \rightarrow (4, 5) \rightarrow (6, 3) \rightarrow 0$ (b) 18 m

IDEA 03 • OTHER REAL-LIFE GRAPHS

- Q15** Dr Eslam • Meaning $m = 1.2$: £1.20 per km; $c = 3$: fixed charge £3
- Q16** Practice • Easy (a) -1.5 (b) the depth falls 1.5 cm every second
- Q17** Practice • Easy (a) about £54 (b) B: 14.3p per kWh is more than A's rate
- Q18** Practice • Medium (a) £10 (b) same at 20 miles; Ed cheaper below, Bill cheaper above 15 litres
- Q19** Practice • Medium (a) 6 seconds (b) about 2 m/s (tangent at 6 s)
- Q20** Practice • Hard (a) 13 m/s (b) 60 mph = 26.7 m/s < 30: yes
- Q21** Practice • Hard (c) about 3.5 m/s² (d) cars have a top speed